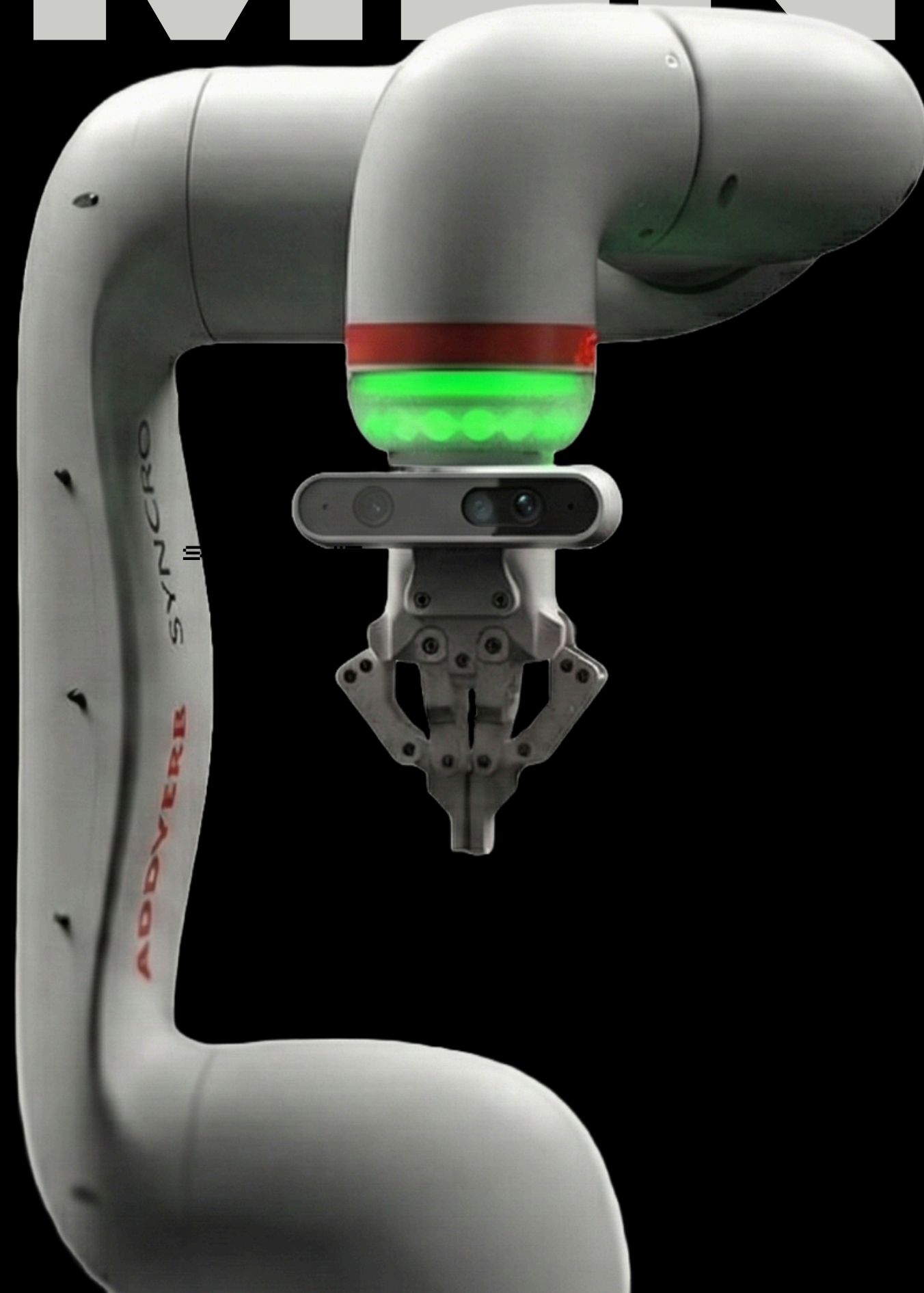
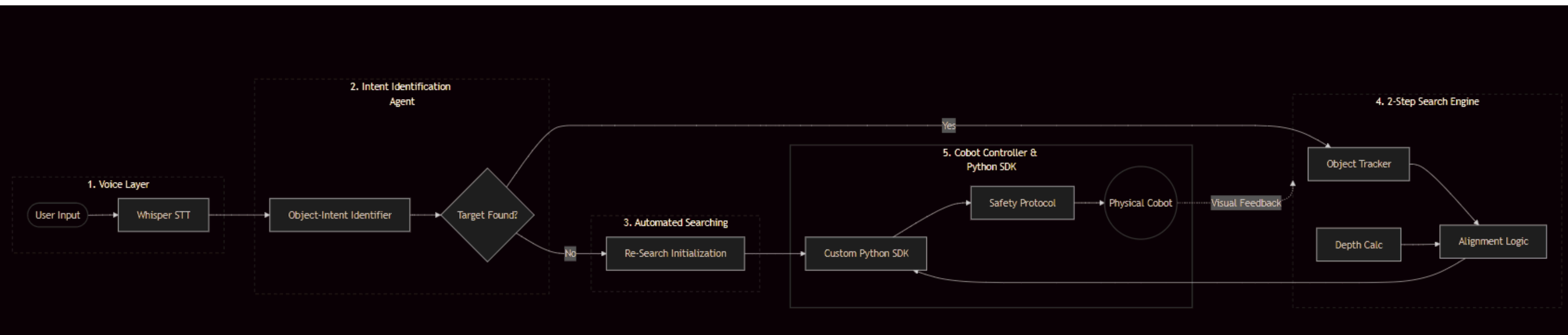


AUMENTA



Aumenta Pipeline



Pipeline Overview

NLP Module

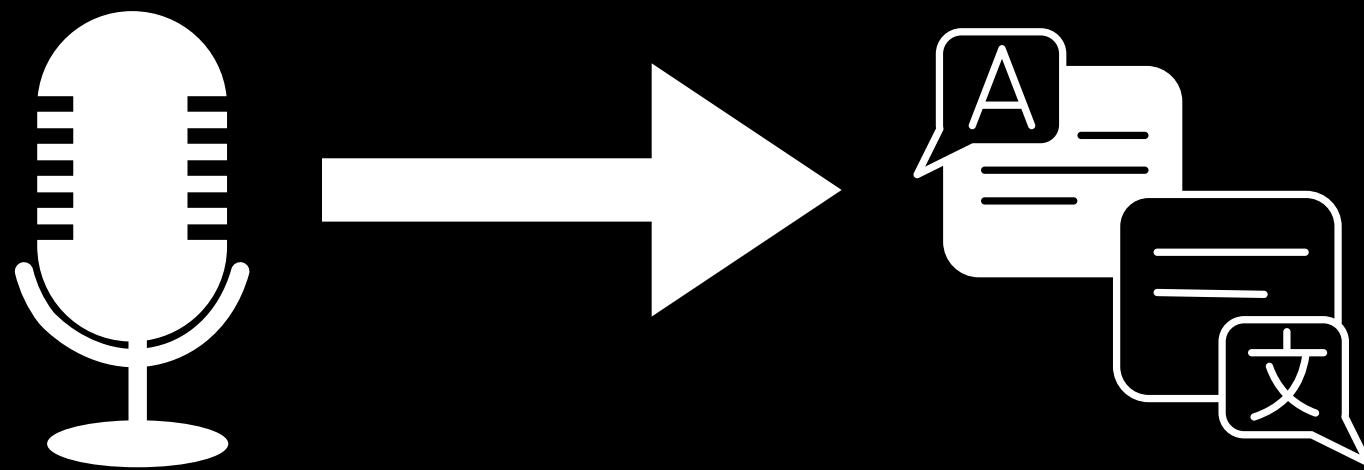
Intent Identification Agent

- **Semantic Filtering:** Instantly filters unstructured speech to remove irrelevant conversational noise.
- **Target Isolation:** Pinpoints the precise object name for immediate vision tracking.

Input: "Pick up the ball"
Target Entity: "ball"



Voice inputs



- Robust Transcription: Accurately converts speech to text using OpenAI's Whisper, even in noisy environments.
- Hands-Free Operation: Enables natural, voice-driven control without the need for manual input.

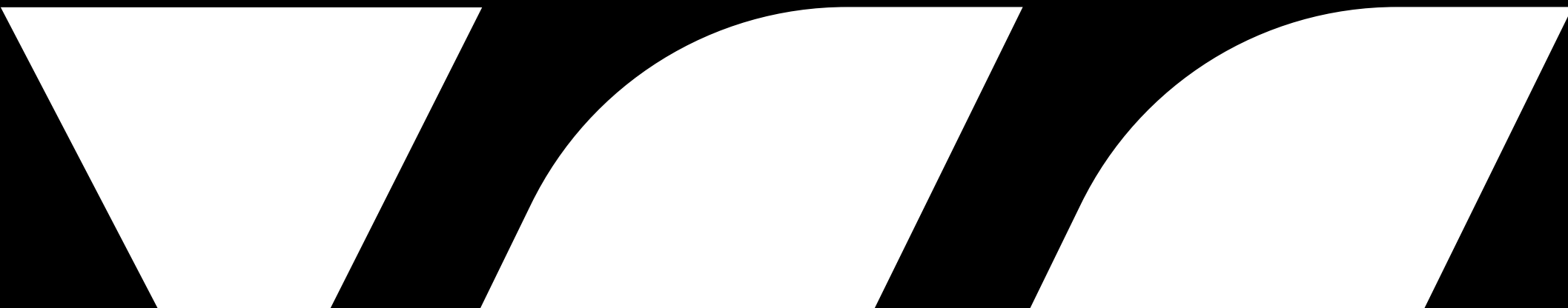
Automated Searching



- **Intelligent Exploration:** Autonomously jogs joints to scan the environment, intelligently searching beyond the camera's static field of view.
- **Operational Necessity:** Critical for robustness, ensuring the system functions even when targets are not initially visible or perfectly placed.
- **Proactive Agency:** Transforms the cobot from a passive observer into an active agent that systematically seeks out its target.

Live Object Detection

- **High-Speed Inference:** Utilizes the lightweight yoloe-26l-seg.pt model to process frames with minimal delay, ensuring the cobot's control loop receives immediate feedback.
- **Dynamic Alignment:** Continuously updates bounding box coordinates at optimized 0.1-second intervals, enabling the control loop to instantly calculate center deviation and correct the cobot's position in real-time.

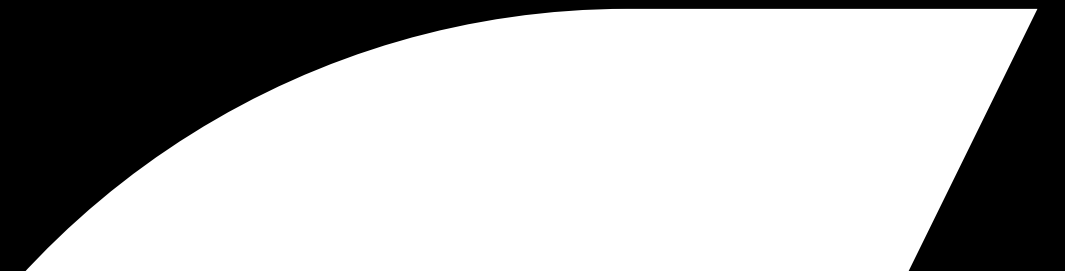
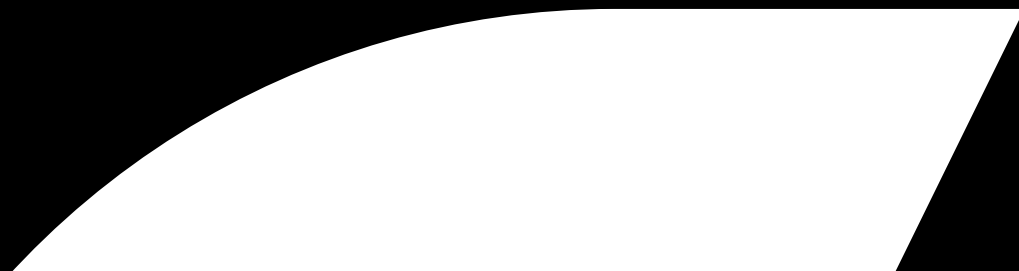


2-Step Engine

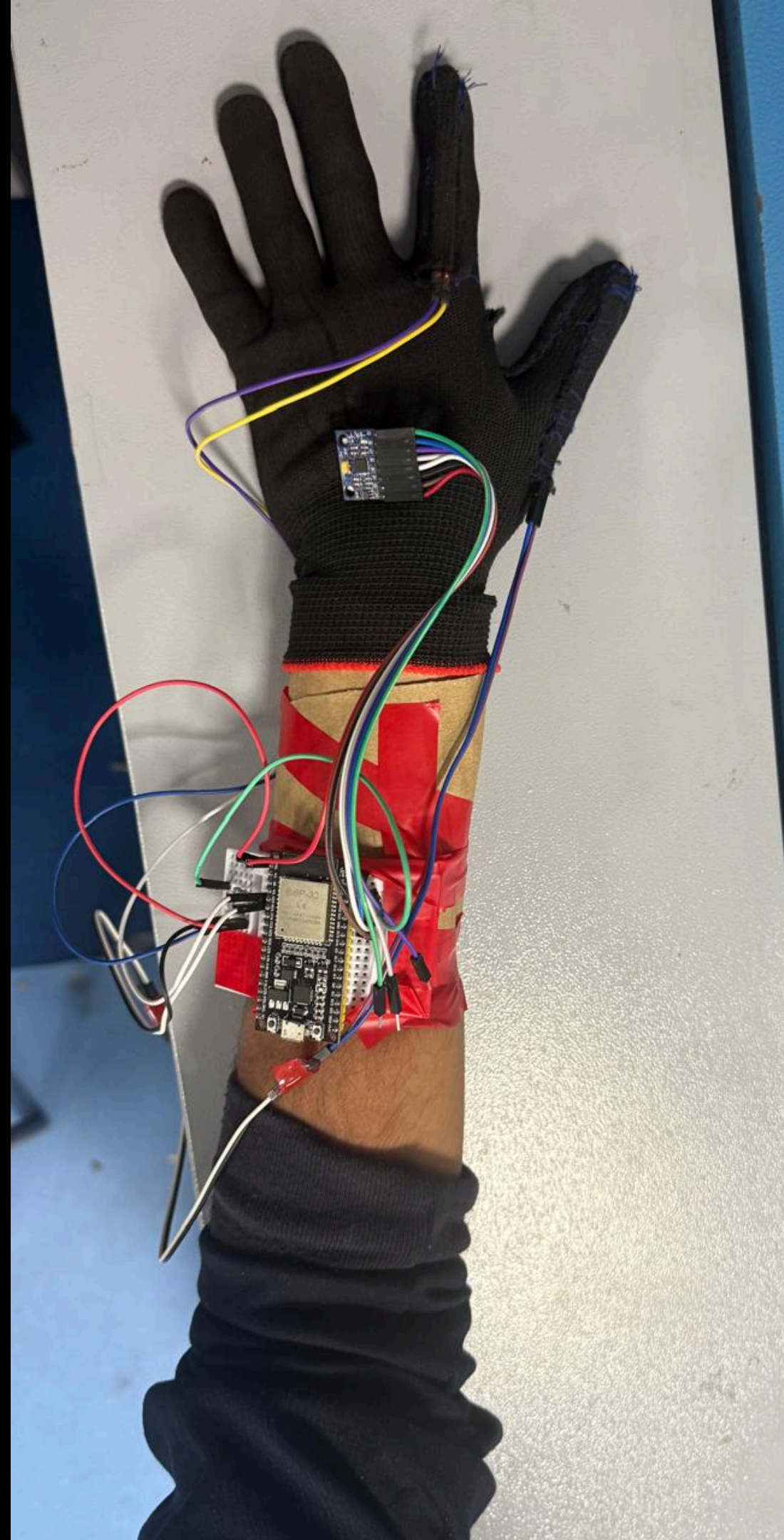
- **X-Y Centering Logic:** First, the system calculates the pixel deviation (Δx , Δy) between the camera's optical center and the object's bounding box center, issuing discrete jog commands (LEFT, RIGHT, UP, DOWN) to align the axes until the error is within a defined threshold.
- **Depth-Based Final Approach:** Once X-Y alignment is locked, the pipeline queries the `pyrealsense2` depth stream at the object's center coordinates (ox , oy) to retrieve the precise Z-distance in meters, triggering the final forward approach or pickup sequence.



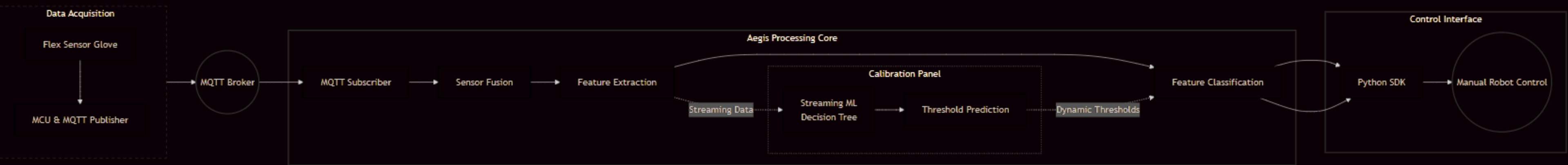
Cobot Controller & Python SDK



Atlas



Atlas Pipeline



Streaming ML

powered by  *River*

Latency

Glove ~ 40 ms

Classification ~ 60 ms

- **Streaming ML for Gesture Detection**
- **Live Training on per user data**
- **Descision Tree based classification**



Classification Metrics

Gesture **Accuracy**

UP	96.23 %
DOWN	98.35 %
LEFT	98.21 %
RIGHT	95.40 %
FORWARDS	99.42 %
BACKWARDS	98.32 %
NEUTRAL	99.98 %
ANGLE (XY)	97.98 %
ANGLE (Z)	97.72 %





THANK YOU

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Sinam J
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Dhananjay Dhumal**

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